Robust Multi-UAV Cooperative Maritime Object Recognition Under Dynamic Aerial Perspectives via Conflict-Modulated Generative Continual Learning Framework

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***Abstract*—Multi-uncrewed aerial vehicle (UAV) cooperative maritime object recognition aims to maintain high accuracy under dynamic aerial perspectives for maritime search and rescue missions. Existing continual learning methods retain knowledge by approximating the global distribution of prior data but fail to address cross-perspective knowledge conflicts caused by distribution shifts across aerial perspectives, leading to gradient perturbations that harm consistency and accuracy in dynamic maritime environments. To address the issues, we propose a conflict-modulated (ConMod) generative continual learning framework, comprising generative perspective-robust conflict estimation and ConMod continual learning modules. The generative perspective-robust conflict estimation employs a perspective-aware scene generator that embeds maritime knowledge priors as perspective constraints to augment the data distribution, thereby facilitating explainable cross-perspective conflict association and promoting robust conflict index estimation. It also incorporates a dual-modal conflict index estimator that integrates geometric distortion and environmental variation branches to estimate conflict indices by associating simulated scenes with perspective-robust data distributions. Furthermore, ConMod continual learning introduces a perspective-specific triplet loss to regularize consistency by aligning geometric and environmental features within perspective-specific representation space. Additionally, a ConMod loss treats conflict indices as modulation weights to identify and reinforce representative conflict experiences from a cross-perspective memory buffer, guiding gradient updates toward global optimization. Results on the SeaDronesSee-CL and SeaDronesSee-CL-v2 datasets show that ConMod effectively improves multi-UAV cooperative maritime object recognition by identifying and mitigating cross-perspective knowledge conflicts.**

***Index Terms*—Conflict modulated (ConMod), generative continual learning, maritime object recognition, maritime search and rescue, multi-uncrewed aerial vehicle (UAV) cooperative perception.**

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I. Introduction

MARITIME search and rescue have become a critical concern in global safety governance due to the high frequency and severity of marine emergencies. According to the European Maritime Safety Agency (EMSA), a total of 26 595 marine casualties and incidents were reported over the past decade, averaging about 2660 cases annually. This highlights the urgent need for effective and intelligent maritime search and rescue systems [1]. Equipped with networked communication and high mobility, maritime uncrewed aerial vehicles (UAVs) provide high-resolution aerial imagery for detailed observation of rescue objects and support real-time operations [2]. Multi-UAV cooperative perception further expands spatial coverage and improves recognition accuracy in dynamic marine environments, overcoming the limitations of individual UAVs, such as restricted views. As a result, multi-UAV cooperative maritime object recognition has emerged as a promising approach for rapid localization and identification of maritime rescue objects from diverse UAV imagery [3], [4].

Inspired by the success of convolutional neural network (CNN) architectures, deep learning has become a dominant approach for maritime UAV object recognition, including one- and two-stage object detection methods [5], [6]. One-stage object detection [7], [8] enables real-time onboard recognition by jointly performing localization and classification across dense candidate regions using lightweight CNNs [9], [10], [11]. Specifically, DCLANet [12] introduces a local attention mechanism to improve recognition performance. Building upon the VTSaR dataset [13], APDNet [14] is formulated as a lightweight distillation framework for edge deployment, incorporating teacher–student knowledge transfer, low-precision quantization, and frame-by-frame tracking. In parallel, VTSaRNet [15] integrates a Union Transformer, bimodal feature fusion, and adaptive channel recalibration to address sparsely distributed objects under complex illumination conditions. Despite the advances in real-time inference, the aforementioned methods still face challenges with small-scale maritime objects and complex maritime environments [16], [17]. To address these issues, two-stage object detection methods [18], [19] handle small-scale maritime objects by generating region proposals, followed by parallel

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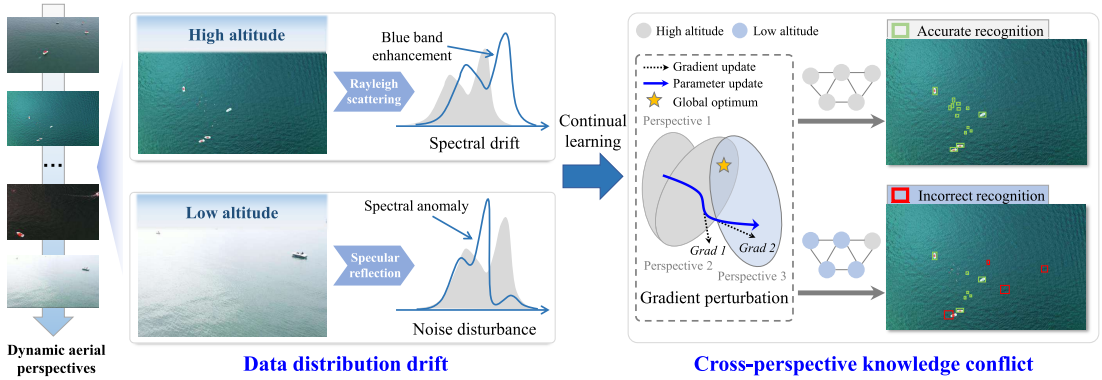


Fig. 1. Cross-perspective knowledge conflicts arise from data distribution drift across dynamic aerial perspectives (e.g., altitude and gimbal pitch angles), hindering the consistency and accuracy of multi-UAV cooperative maritime object recognition.

heads for classification and bounding box regression. R-CNN [20] applies region-based feature encoding to extract deep CNN features from multi-UAV aerial imagery, while Faster R-CNN [21], [22] extends R-CNN by incorporating a region proposal network (RPN) for end-to-end proposal generation and multiscale feature fusion designed to enable accurate recognition of small-scale maritime objects.

Despite progress in deep learning-based maritime UAV object recognition, most methods rely on fixed aerial perspectives, assuming predefined flight trajectories with static sensor configurations, which limits their applicability in multi-UAV cooperative maritime object recognition. As a result, models under this paradigm are trained on static datasets with fixed parameters during inference. To address this limitation, recent multi-UAV cooperative methods have advanced robustness by leveraging both decision- and feature-level fusion strategies to adapt to multi-UAV perspectives [23], [24], [25]. Specifically, MIA-Net [26] adopts a decision-level fusion strategy via local–global matching to achieve cross-UAV object association and handle occlusions, while TCFNet [24] and TSMMT [27] introduce feature-level fusion by estimating cross-UAV transformation relationships and incorporating temporal–spatial feedback loops to enhance cooperative capability in complex scenes. Although these approaches provide valuable insights into multi-UAV multiperspective integration, they remain limited in scenarios that require continual adaptation to dynamic aerial perspectives. This limitation poses a challenge for maritime search and rescue missions, which inherently demand UAVs to adjust parameters such as altitude and gimbal pitch angles to maintain optimal observation conditions. Therefore, there is an urgent need for robust solutions that enable models to continually adapt to variations in dynamic aerial perspectives, ensuring stable performance across diverse maritime scenarios.

Recent advances in continual learning paradigm [28], [29], [30] have made it feasible to formulate multi-UAV cooperative maritime object recognition as a domain-incremental learning task, where models sequentially accumulate knowledge from diverse aerial perspectives while preserving previously learned information. Representative approaches include parameter regularization, functional regularization, and replay methods [30], [31]. Parameter regularization constrains updates to important parameters to reduce catastrophic forgetting [32], [33]. For example, DILRS [28] introduces domain-residual adapters to address label space shifts caused by sensor heterogeneity, enabling cross-platform adaptation. However, such methods fail to accurately identify important parameters within highly dynamic environments. Functional regularization explicitly maintains prior knowledge by enforcing consistency in the input–output mapping across sequential tasks [34]. DENet [31] employs dynamic equilibrium constraints and self-distillation to improve generalization across heterogeneous UAV domains. Structured inheritance [35] further incorporates hierarchical feature propagation and spatial constraints to preserve learned spatial dependencies. Yet, these methods cannot achieve long-term knowledge accumulation, as they are inherently ill-equipped to handle the nonstationary distributions arising from heterogeneous task sequences. Replay-based methods address this by retaining representative experiences in external memory or generative models [36], [37]. For instance, IDHT [38] develops an incremental spectral detector to align feature spaces from different UAV sensors, while LAPNet [30] introduces layout-aware prioritizing learning to preserve representative features and mitigate cross-sensor knowledge forgetting. Deep generative replay methods employ generative models such as stable diffusion [39], [40] to synthesize high-fidelity remote sensing imagery, which serves as replay experiences to approximate historical data distributions without retaining raw data.

Although existing continual learning methods effectively preserve knowledge by simulating the global distribution of historical experiences, multi-UAV cooperative maritime object recognition remains unresolved due to cross-perspective knowledge conflicts. Specifically, these methods fail to capture inherent conflict associations induced by data distribution drift across dynamic aerial perspectives, which degrade the consistency and accuracy in dynamic maritime environments (Fig. 1). This requires explainable modeling capabilities, capable of explicitly identifying the underlying sources of cross-perspective knowledge conflicts and providing a theoretical basis for understanding how UAV perspective variations influence cross-perspective conflicts. In particular, high-altitude UAV perspectives are affected by increased atmospheric absorption, which induces Rayleigh scattering and

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enhances spectral responses in the blue band. In contrast, low-altitude UAV perspectives introduce strong specular reflection, resulting in localized noise disturbances and abnormal spectral characteristics. These discrepancies induce biased parameter updates and gradient perturbations, resulting in convergence instability and catastrophic forgetting, which ultimately hinder reliable maritime object recognition across perspectives.

To address these issues, this article proposes a ConMod framework for multi-UAV cooperative maritime object recognition. It investigates a conflict modulation mechanism to identify and mitigate cross-perspective knowledge conflicts, offering solutions for conflict management across dynamic aerial perspectives. The main contributions are as follows.

1) To alleviate cross-perspective knowledge conflicts in multi-UAV cooperative maritime object recognition, we propose ConMod, a ConMod generative continual learning framework with two modules. The generative perspective-robust conflict estimation module embeds maritime knowledge priors into a dual-modal conflict index estimator to accurately quantify cross-perspective conflicts. The ConMod continual learning module enforces consistency constraints to regularize perspective-specific representations and leverages conflict indices to reinforce representative conflict experiences while adaptively modulating gradient updates for global optimization.
2) A generative perspective-robust conflict estimation module adopts generative image editing with controllable instruction priors to estimate cross-perspective knowledge conflicts. It employs a perspective-aware scene generator that models probabilistic state transitions across perspectives and provides perspective-constrained image editing instructions, embedding maritime knowledge priors into simulated maritime scenes, enabling explainable conflict association and improving robustness in handling anomalous cross-perspective distribution shifts. The module integrates a dual-modal conflict index estimator quantifying conflicts by associating simulated scenes with perspective-robust data distributions. It encodes dual-modal features from a geometric distortion branch based on UAV perspective metadata and an environmental variation branch driven by simulated maritime scenes, estimating conflict indices via regularized Euclidean distances to cluster centroids in both geometric and environmental feature spaces.
3) A ConMod continual learning module adaptively modulates gradient updates using conflict indices to facilitate global parameter optimization. It introduces a perspective-specific triplet loss to enforce dual-modal alignment between geometric and environmental features, with the environmental feature as the anchor, the matching geometric feature as the positive, and a geometric feature from another UAV perspective as the negative. This loss enhances perspective-specific feature representations by reducing intraperspective variation while maximizing interperspective separation. A ConMod loss employs conflict indices as modulation weights to identify and reinforce representative conflict experiences stored in a cross-perspective memory buffer, enabling the learning of systematic distribution shifts in cross-perspective maritime data and guiding gradient updates toward global optimality.

To evaluate the proposed ConMod framework for multi-UAV cooperative maritime object recognition, we constructed two benchmark datasets, comprising 26 670 objects from seven UAV perspectives across six categories. The rest of this article is organized as follows. Section II describes the proposed ConMod framework in detail, while Sections III and IV presents the experimental results and ablation analysis. Finally, Section V concludes this article.

II. ConMod: Multi-UAV Cooperative Maritime Object Recognition Framework

This section presents the ConMod framework for multi-UAV cooperative maritime object recognition, as shown in Fig. 2. ConMod consists of two main modules: generative perspective-robust conflict estimation, which measures knowledge conflicts caused by cross-perspective inconsistencies, and ConMod continual learning, which learns systematic distribution shift patterns in multiperspective maritime data. In the generative perspective-robust conflict estimation module, the original image along with UAV perspective metadata is fed into a perspective-aware scene generator. This generator explicitly produces diverse maritime scene data to simulate cross-perspective conflict association, enhancing the model's robustness against unexpected shifts in anomalous data distributions. The UAV perspective metadata and simulated images are then utilized for conflict index estimation, which is decomposed into two branches: a geometric distortion branch and an environmental variation branch. In the ConMod continual learning module, a perspective-specific triplet loss is employed to enforce alignment between geometric and environmental features derived from the respective branches. Additionally, a ConMod loss assigns conflict indices as modulation weights to identify and reinforce representative conflict experiences to enable globally optimal parameter optimization.

A. Generative Perspective-Robust Conflict Estimation

The generative perspective-robust conflict estimation module incorporates generative image editing that leverages controllable instruction priors to estimate cross-perspective knowledge conflicts. It employs a perspective-aware scene generator that takes the original image and UAV perspective metadata as input to simulate maritime scenarios, establishing explainable conflict associations to improve the model's robustness to unexpected anomalies under cross-perspective scenarios. In addition, a dual-modal conflict index estimator incorporates geometric distortion and environmental variation to estimate potential knowledge conflicts.

1) Perspective-Aware Scene Generator: To simulate the distribution shifts across UAV perspectives in maritime data, a perspective-aware scene generator is proposed (Fig. 3). This generator takes the original image I_0 with the spectral wavelength λ and UAV perspective metadata (e.g., altitude h and angle θ) as inputs. It employs the perspective-constrained

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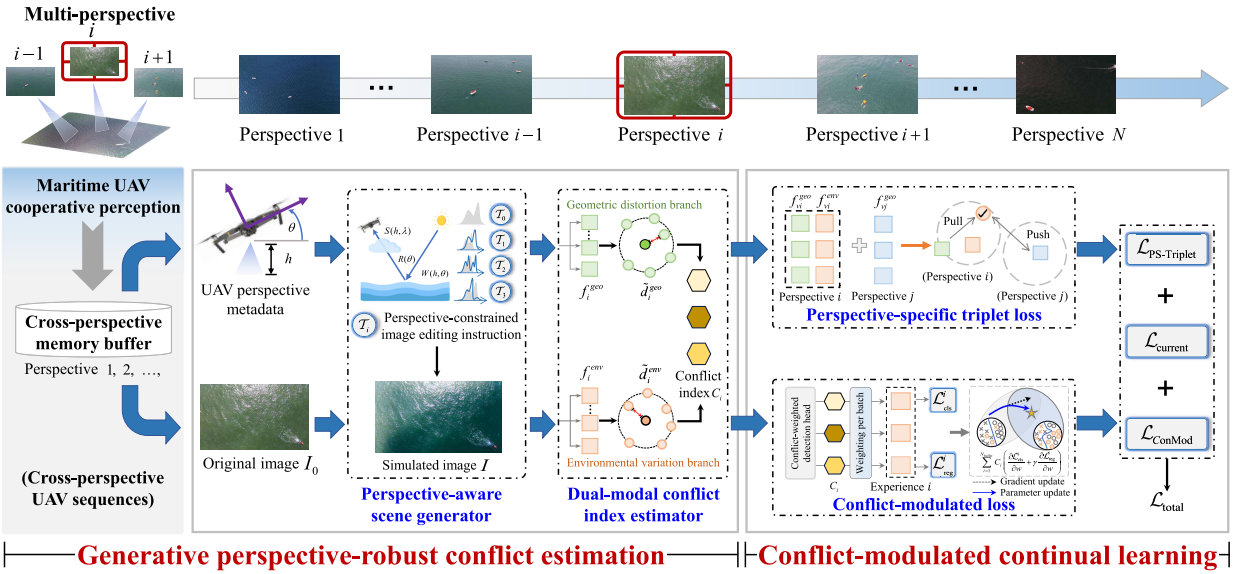


Fig. 2. ConMod comprises a generative perspective-robust conflict estimation module to quantify cross-perspective knowledge conflicts and a ConMod continual learning module to modulate gradient updates for global optimization.

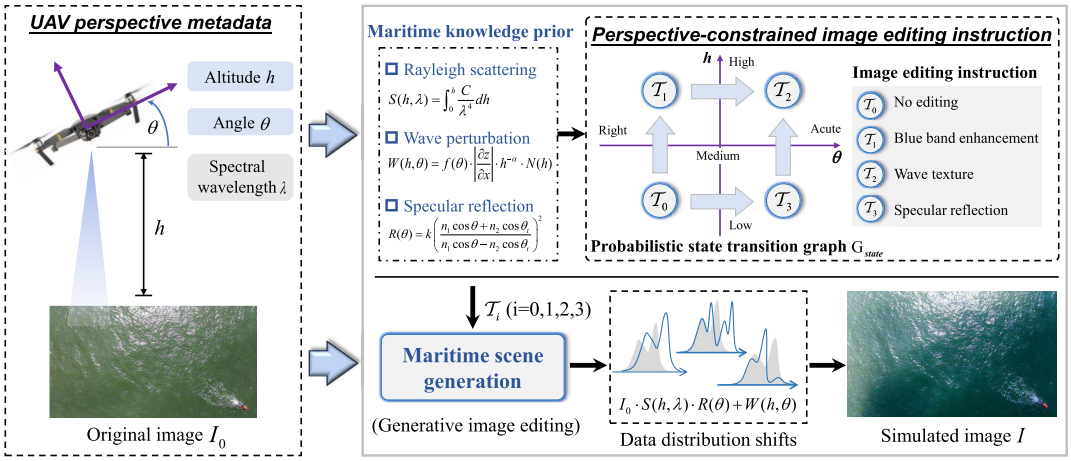


Fig. 3. Perspective-aware scene generator simulates maritime scenes by applying perspective-constrained image editing instructions guided by maritime knowledge priors such as Rayleigh scattering, wave perturbation, and specular reflection.

image editing instruction to simulate maritime scenes by incorporating maritime knowledge priors, including Rayleigh scattering $S(\cdot)$, wave perturbation $W(\cdot)$, and specular reflection $R(\cdot)$, which represent the underlying sources of knowledge conflicts. The simulated image I is formulated as

$$I(h,\theta,\lambda) = I_0 \cdot S(h,\lambda) \cdot R(\theta) + W(h,\theta) \tag{1}$$

where the Rayleigh scattering $S(\cdot)$ and specular reflection $R(\cdot)$ are related to I_0 through a multiplicative relationship, while the wave perturbation $W(\cdot)$ acts linearly on I_0. The three maritime knowledge priors are formally defined as

$$S(h,\lambda) = \int_0^h \frac{C}{\lambda^4}\,\mathrm{dh} \tag{2}$$

$$W(h,\theta) = f(\theta) \cdot \left|\frac{\partial z}{\partial x}\right| \cdot h^{-\alpha} \cdot N(h) \tag{3}$$

$$R(\theta) = k \cdot G(\theta,\theta_i)\left(\frac{n_1\cos\theta_i - n_2\cos\theta_t}{n_1\cos\theta_i + n_2\cos\theta_t}\right)^2 \tag{4}$$

where C is an environmental constant related to Rayleigh scattering, $f(\theta)$ is an angular modulation factor, $(\partial z/\partial x)$ represents the local wave slope, α is an attenuation coefficient, $N(h)$ indicates the number of visible waveforms at altitude h, k is a scaling constant for specular reflection, $G(\theta,\theta_i)$ is a Gaussian weighting function modeling the alignment between the angle θ and the incidence angle θ_i, θ_t is the angles of refraction, and n_1 and n_2 are the refractive indices of air and water.

The perspective-constrained image editing instruction integrates the three aforementioned maritime knowledge priors into maritime scene generation. It utilizes a predefined probabilistic state transition graph G_{state} to quantify the transformation patterns between different perspectives. This graph captures the relationship between UAV perspective variations and corresponding data distribution shifts, thus establishing a structured representation for the explainability of conflict associations. The graph G_{state} is constructed based on variations in altitude h and angle θ, providing semantic descriptions that

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guide generative models for controllable image editing. Specifically, InstructDiffusion [41], a generative image editing model guided by textual instructions, is employed for maritime scene generation. Built upon the Stable Diffusion model, InstructDiffusion introduces instruction-based fine-tuning, enabling more precise and controllable image manipulation guided by textual prompts. The perspective-constrained image editing instructions include four types: no editing $\mathcal{T}_0$, blue band enhancement $\mathcal{T}_1$, wave texture enhancement $\mathcal{T}_2$, and specular reflection enhancement $\mathcal{T}_3$. The state transition rule $\mathcal{T}(h,\theta)$ is applied to determine the instruction $T_i(i = 0,1,2,3)$ by comparing the altitude h with the medium altitude h_M and the angle θ with the medium angle θ_M, which is expressed as follows:

$$\mathcal{T}(h,\theta) = \begin{cases} \mathcal{T}_0, & h > h_M \quad \text{and } \theta > \theta_M \\ \{\mathcal{T}_0, \mathcal{T}_1\}_p, & h = h_M \quad \text{and } \theta \le \theta_M \\ \{\mathcal{T}_0, \mathcal{T}_2\}_p, & (h > h_M \text{and } \theta \le \theta_M) \\ & \text{or } (h \le h_M \text{and } \theta > \theta_M) \\ \{\mathcal{T}_0, \mathcal{T}_1, \mathcal{T}_3\}_p, & h < h_M \quad \text{and } \theta \le \theta_M \end{cases} \tag{5}$$

where p denotes the uniform probability of element selection from the set $\{\cdot\}$, incorporating the stochasticity to enhance data distribution generalization.

Formally, blue band enhancement $\mathcal{T}_1$ simulates Rayleigh scattering $S(h,\lambda)$, where higher UAV altitude h strengthens the response at shorter wavelengths λ. This process is related to radiation distortion, as shorter wavelengths such as the blue band are highly sensitive to atmospheric scattering in maritime scenes. The generative image editing is guided by the instruction: "enhance the blue band in the current aerial view over the sea." Wave texture $\mathcal{T}_2$ models wave perturbation $W(h,\theta)$, where a higher altitude increases wave count but reduces individual wave detail, while a larger observation angle θ enhances high-frequency textures, increasing visual complexity. The corresponding instruction is: "enhance the wave textures to increase the complexity of the background." Specular reflection $\mathcal{T}_3$ leverages Fresnel reflection law to simulate glints arising from increased UAV observation angles θ. The generative image editing is guided by the instruction: "enhance the specular reflection to simulate sunlight on the water surface." These instructions collectively characterize the influence of UAV perspective variations on maritime scene appearance, serving as physics-informed priors to guide generative image editing for explainable conflict association that enhances the model's robustness against unexpected anomalies in cross-perspective scenarios.

2) Dual-Modal Conflict Index Estimator: The dual-modal conflict index estimator (Fig. 4) employs UAV perspective geometric priors to estimate a conflict index by modeling both visual appearance differences and UAV perspective geometric distortions, enabling robust and explainable conflict estimation. Specifically, the conflict index consists of two conflict branches: geometric distortion and environmental variation. Each branch encodes the respective feature representations, i.e., geometric features $\mathcal{F}_{\text{geo}}$ and environmental features $\mathcal{F}_{\text{env}}$, and quantifies the deviation of individual feature vectors by employing a regularized Euclidean feature distance metric. This metric measures the dispersion of each feature vector

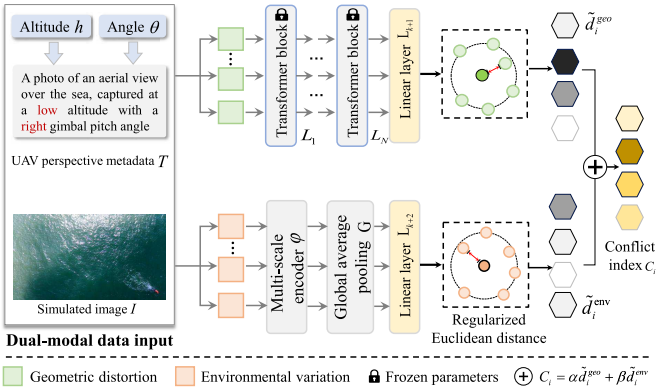


Fig. 4. Dual-modal conflict index estimator estimates the conflict index by using a regularized Euclidean feature distance metric within geometric and environmental feature spaces.

from the corresponding cluster centroid, capturing relative inconsistency within geometric and environmental features to eliminate the intrinsic cross-modal distribution gap.

The geometric distortion branch leverages UAV perspective metadata (i.e., altitude h and angle θ) for geometric feature representation. This information is encoded using a text encoder Φ_{text} consisting of multiple Transformer blocks with frozen parameters, retaining prior knowledge from large-scale image–text pairs and generating general features that capture geometric distortion variations. A linear layer $\mathcal{L}_{k+1}$ then generates geometric features $\mathcal{F}_{\text{geo}}$ by reducing feature dimensionality from $d_l = 512$ to $d_v = 64$, aggregating key channel information and encoding cross-perspective geometric distortions effectively. $\mathcal{F}_{\text{geo}}$ can be expressed as

$$\mathcal{F}_{\text{geo}} = \Phi_{\text{text}}(h,\theta) = \mathcal{L}_{k+1}\left(L^{(N)} \circ, \ldots, \circ L^{(1)}(T(h,\theta))\right) \tag{6}$$

where T represents the UAV perspective text template, such as `A photo of an aerial view over the sea, captured at a` $\langle h \rangle$ `altitude with` $\langle \theta \rangle$ `gimbal pitch angle`, providing a structured textural description. $L^{(n)}$ represents the nth Transformer block, and $\mathcal{L}_{k+1}$ is the linear layer that reduces the dimensionality. This encodes geometric distortion into geometric features that associate perspective variations for explainable conflict estimation.

The environmental variation branch takes the simulated image I as input and constructs an image encoder Φ_{img} to extract environmental features. The encoder Φ_{img} includes a ResNet backbone $\varphi(\cdot)$ to extract multiscale spatial features, enabling robust environmental perception, which is followed by global average pooling $\mathcal{G}(\cdot)$ to achieve translation-invariant environmental feature representation while retaining global contextual information. A linear layer $\mathcal{L}_{k+2}$ is then applied to project the high-dimensional pooled features of dimension d_{l2} to environmental features $\mathcal{F}_{\text{env}}$ of dimension d_{v2}, where $d_{l2} = 256$ and $d_{v2} = 64$. Φ_{img} is expressed as

$$\mathcal{F}_{\text{env}} = \mathcal{L}_{k+2}(\mathcal{G}(\varphi(I))). \tag{7}$$

After obtaining the geometric and environmental features, the conflict index is estimated by determining the cluster centroids of each feature distribution. Given geometric and environmental feature sets $\mathcal{F}_{\text{geo}} = \{f_i^{\text{geo}}\}_{i=1}^{N}$ and

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$\mathcal{F}_{\text{env}} = \{f_i^{\text{env}}\}_{i=1}^{N}$, the cluster centroids c^{geo} and c^{env} of these feature distributions can be defined as

$$c^{\text{geo}} = \frac{1}{N}\sum_{i=1}^{N} f_i^{\text{geo}}, \quad c^{\text{env}} = \frac{1}{N}\sum_{i=1}^{N} f_i^{\text{env}}. \tag{8}$$

The cross-perspective inconsistency of each feature vector from the corresponding cluster centroid is measured using a regularized Euclidean distance

$$d_i^{\text{geo}} = \|f_i^{\text{geo}} - c^{\text{geo}}\|_2 \quad d_i^{\text{env}} = \|f_i^{\text{env}} - c^{\text{env}}\|_2 \tag{9}$$

$$\tilde{d}_i^{\text{geo}} = \frac{d_i^{\text{geo}} - \min\left(d^{\text{geo}}\right)}{\max\left(d^{\text{geo}}\right) - \min\left(d^{\text{geo}}\right)} \tag{10}$$

$$\tilde{d}_i^{\text{env}} = \frac{d_i^{\text{env}} - \min\left(d^{\text{env}}\right)}{\max\left(d^{\text{env}}\right) - \min\left(d^{\text{env}}\right)} \tag{11}$$

where $\|\cdot\|_2$ denotes the Euclidean distance and d_i^{geo} and d_i^{env} are normalized to $\tilde{d}_i^{\text{geo}}$ and $\tilde{d}_i^{\text{env}}$ within the range of $[0, 1]$ using min–max normalization, respectively.

The conflict index C_i quantifies the deviation of each feature vector from the corresponding cluster centroid, defined as

$$C_i = \alpha\tilde{d}_i^{\text{geo}} + \beta\tilde{d}_i^{\text{env}} \tag{12}$$

where α and β are weighting coefficients that balance the impact of geometric distortion and environmental variation, ensuring robust conflict estimation by mitigating noise from visual variations. By default, α and β are set to 0.5.

B. ConMod Continual Learning

To adaptively modulate gradient backpropagation for globally optimal parameter updates, we propose a ConMod continual learning module adopting conflict indices as modulation weights to identify and reinforce representative conflict experiences in a cross-perspective memory buffer. Specifically, this module incorporates a perspective-specific triplet loss to align geometric and environmental features, while a ConMod loss leverages the conflict index to guide gradient modulation across UAV perspectives. The total loss $\mathcal{L}_{\text{total}}$ includes the perspective-specific triplet loss $\mathcal{L}_{\text{PS-Triplet}}$, the ConMod loss $\mathcal{L}_{\text{ConMod}}$ for the memory experiences across UAV perspectives, and $\mathcal{L}_{\text{current}}$ for the current perspective experiences. The total loss is defined as

$$\mathcal{L}_{\text{total}} = \lambda_{\text{PS}}\mathcal{L}_{\text{PS-Triplet}} + \left(1 - \frac{1}{N_{\text{perspective}}}\right)\mathcal{L}_{\text{ConMod}} + \frac{1}{N_{\text{perspective}}}\mathcal{L}_{\text{current}} \tag{13}$$

where $N_{\text{perspective}}$ represents the number of UAV perspectives encountered and λ_{PS} is the regularization parameter of the perspective-specific triplet loss, set to 0.5 by default.

1) Perspective-Specific Triplet Loss: To ensure dual-modal feature consistency, a perspective-specific triplet loss is introduced to enforce alignment between geometric and environmental features. Traditional triplet loss enforces general feature discrimination by separating arbitrary positive and negative pairs but lacks sensitivity to UAV perspective variations. In contrast, the proposed loss integrates perspective constraints as prior knowledge and constructs positive and negative samples based on UAV perspectives. Specifically, for each feature pair from the geometric and environmental branches, the environmental feature serves as the anchor, the corresponding geometric feature as the positive sample, and a randomly selected geometric feature from a different UAV perspective as the negative sample. This strategy ensures that negative samples are drawn from all available perspectives, mitigating overfitting to a limited set of hard negatives. This loss function minimizes intraperspective differences while maximizing inter-perspective separation, effectively correcting outlier anchors to impose consistency regularization on perspective-specific representations. The perspective-specific triplet loss $\mathcal{L}_{\text{PS-Triplet}}$ is formulated as follows:

$$\mathcal{L}_{\text{PS-Triplet}} = \frac{1}{N}\sum_{i=1}^{N}\max\left(\|f_{vi}^{\text{env}} - f_{vi}^{\text{geo}}\|_2 - \|f_{vi}^{\text{env}} - f_{vj}^{\text{geo}}\|_2 + m,\ 0\right) \tag{14}$$

where f_{vi}^{geo} and f_{vi}^{env} represent the geometric and environmental features extracted from the same UAV perspective i, serving as positive samples. f_{vj}^{geo} is the geometric feature randomly selected from a different UAV perspective j as the negative sample at each iteration to enhance generalization. The margin m is introduced to enforce a separation between positive and negative samples, while N is the batch size.

2) ConMod Loss: The ConMod loss (Fig. 5) treats conflict indices as modulation weights to learn systematic distribution shift patterns in cross-perspective maritime data. Specifically, the ConMod loss $\mathcal{L}_{\text{ConMod}}$ utilizes the conflict index C_i to quantify discrepancies across cross-perspective data distributions, guiding the conflict-weighted detection head that includes a classification head and a box regression head to effectively incorporate conflict features. Given the conflict index C_i as input, the ConMod loss $\mathcal{L}_{\text{ConMod}}$ is defined as

$$\mathcal{L}_{\text{ConMod}} = \sum_{i=1}^{N_{\text{buffer}}}\left[\mathcal{L}_{\text{cls}}\left(C_i, \hat{y}_i, y_i\right) + \gamma p_i^{*}\mathcal{L}_{\text{reg}}\left(C_i, \hat{b}_i, b_i\right)\right] \tag{15}$$

where N_{buffer} represents the number of memory experiences in the cross-perspective memory buffer and $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{reg}}$ denote the losses for the classification head and the box regression head, respectively. γ is a weight coefficient between the two heads, with a default value of 1. $\hat{y}_i$ and y_i represent the predicted label and the ground truth label, respectively, while

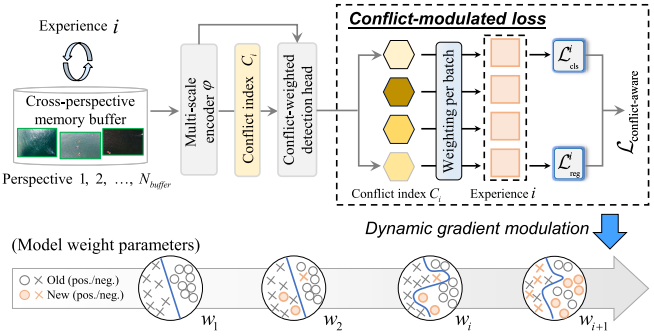


Fig. 5. ConMod loss leverages the conflict index to construct a conflict-weighted detection head, enabling gradient modulation for globally optimal parameter updates.

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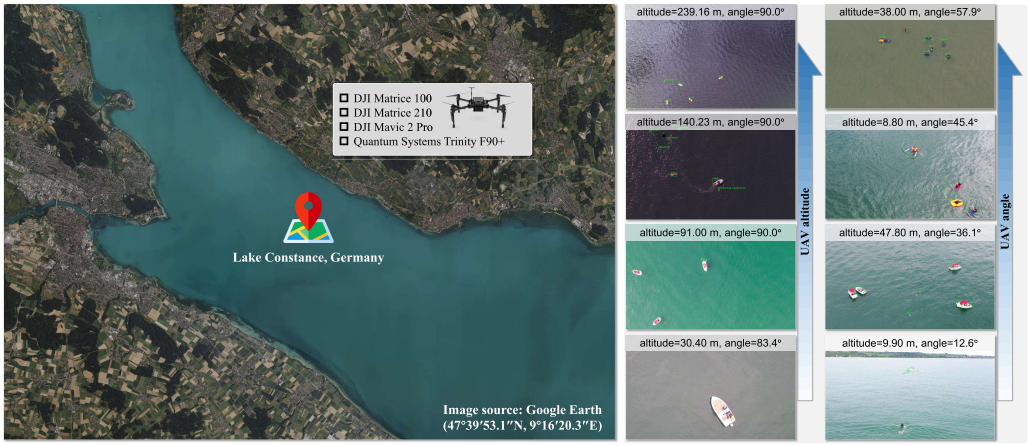


Fig. 6. SeaDronesSee-CL and SeaDronesSee-CL-v2 datasets were collected at Lake Constance, Germany, with UAV platforms, including the DJI Matrice 100, Matrice 210, Mavic 2 Pro, and Trinity F90 +, under varying altitudes and angles.

$\hat{b}_i$ and b_i represent the predicted and true bounding boxes, respectively. p_i^* represents an indicator where $p_i^* = 1$ indicates a positive sample and $p_i^* = 0$ indicates a negative sample. The expressions for $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{reg}}$ are as follows:

$$\mathcal{L}_{\text{cls}} = -C_i \cdot \sum_{k=1}^{K} y_{i,k} \log \frac{e^{\hat{y}_{i,k}}}{\sum_j e^{\hat{y}_{i,j}}} \tag{16}$$

$$\mathcal{L}_{\text{reg}} = C_i \cdot \sum_{m\in\{x,y,w,h\}} \text{SmoothL1}\left(\hat{b}_{i,m} - b_{i,m}\right) \tag{17}$$

where K represents the number of classes and m includes the center coordinates (x, y) and sizes (w, h) of the bounding box. SmoothL1 refers to the Smooth $L1$ loss, which provides a balance between $L1$ and $L2$ losses.

The ConMod continual learning mechanism dynamically guides gradient updates. Specifically, this loss adopts conflict indices as modulation weights to balance the classification loss $\mathcal{L}_{\text{cls}}^i$ and regression loss $\mathcal{L}_{\text{reg}}^i$ for each experience i. Experiences with high conflict indices receive large update magnitudes during parameter optimization, ensuring that cross-perspective inconsistencies are explicitly embedded into the training process for robust cross-perspective continual learning. This mechanism enhances gradient sensitivity to perspective-induced anomalies, allowing the model to prioritize learning from challenging perspectives with high conflict indices. This handles unexpected shifts in anomalous data distributions and guides the model's decision boundary toward a globally optimal solution for multiperspective maritime data. The gradient is expressed as

$$\frac{\partial \mathcal{L}_{\text{ConMod}}}{\partial w} = \sum_{i=1}^{N_{\text{buffer}}} C_i \left(\frac{\partial \mathcal{L}_{\text{cls}}^i}{\partial w} + \gamma \frac{\partial \mathcal{L}_{\text{reg}}^i}{\partial w} \right) \tag{18}$$

where w denotes the model weight parameters and $\mathcal{L}_{\text{cls}}^i$ and $\mathcal{L}_{\text{reg}}^i$ represent the classification and box regression losses of the ith experience, respectively.

III. Experiments

In this section, we first introduce the two benchmark datasets and the evaluation metrics used in the experiments. Then, we present the implementation details and evaluate the performance of ConMod for multi-UAV cooperative maritime object recognition based on the two benchmark datasets.

TABLE I
SeaDronesSee-CL Dataset Contains 4292 Images Captured From Seven Different Perspectives

Task	Perspective	Altitude (m)	Angle (°)	Train	Test	Total
1	HR	157–259	73–90	131	18	149
2	MR	55–157	73–90	529	56	585
3	LR	5–55	73–90	206	58	264
4	LMR	5–55	56–73	169	34	203
5	LM	5–55	40–56	868	128	996
6	LAM	5–55	23–40	900	150	1050
7	LA	5–55	0–23	900	145	1045
Total				3703	589	4292

A. Dataset Descriptions and Evaluation Metrics

To evaluate the performance of multi-UAV cooperative maritime object recognition algorithms, we constructed the SeaDronesSee-CL and SeaDronesSee-CL-v2 by reorganizing the object detection and multispectral object detection tracks of the SeaDronesSee dataset [42], using RGB bands mounted on UAV platforms. As exhibited in Fig. 6, the data acquisition was conducted from August 25 to August 30, 2020, at Lake Constance, Germany. The primary distinction between the two benchmark datasets lies in UAV perspectives. SeaDronesSee-CL was collected at multiple altitudes and angles, while SeaDronesSee-CL-v2 was captured from nadir perspectives (gimbal pitch angle of 90°) at multiple altitudes.

The SeaDronesSee-CL dataset comprises 4292 image patches, each resized to a resolution of 1080 × 1920 pixels (Table I). The dataset features UAV sequences captured from seven representative aerial perspectives. Each is denoted by abbreviations in which the first letter indicates flight altitude (H: high, M: medium, and L: low), and the subsequent letter(s) denote the observation angle (A: acute and R: right). Specifically, the perspectives include high altitude, right angle (HR), medium altitude, right angle (MR), low altitude, right

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TABLE II

SEADRONESSEE-CL DATASET CONTAINS A TOTAL OF 25 087 ANNOTATED OBJECTS ACROSS FIVE CATEGORIES: SWIMMER, BOAT, JET SKI, LIFE-SAVING APPLIANCES, AND BUOY

Class Name	Train	Test	Total
Swimmer	12794	1641	14435
Boat	6158	879	7037
Jet ski	968	88	1056
Life-saving appliances	302	5	307
Buoy	2088	164	2252
Total	**22310**	**2777**	**25087**

TABLE III

SEADRONESSEE-CL-V2 DATASET CONTAINS 1583 OBJECTS ACROSS THREE CATEGORIES: SWIMMER, FLOATER, AND BOAT

Class Name	Train	Test	Total
Swimmer	61	32	93
Floater	388	225	613
Boat	598	279	877
Total	**1047**	**536**	**1583**

angle (LR), low altitude, medium-right angle (LMR), low altitude, medium angle (LM), low altitude, acute-medium angle (LAM), and low altitude, acute angle (LA). The flight altitudes range from 5 to 259 m, with observation angles varying from 0° to 90°.

Table II lists 25 087 annotated maritime objects in the dataset across five classes: swimmer, boat, jet ski, life-saving appliance, and buoy. The "swimmer" class accounts for the largest proportion, with 14 435 instances, as swimmers are a critical object in maritime search and rescue operations.

Table III summarizes 1583 maritime objects in the SeaDronesSee-CL-v2 dataset, categorized into three classes: swimmer, floater (swimmers wearing life jackets), and boat. This dataset serves as a benchmark for small maritime object recognition under nadir perspectives at multiple altitudes (HR, MR, and LR), enabling comprehensive evaluation across diverse maritime search and rescue scenarios.

To evaluate the performance of maritime object recognition, the object detection metric of mean average precision (mAP) is adopted. The mAP measures the tradeoff between precision and recall, averaged across all object classes. It is calculated as the area under the precision–recall (P–R) curve and normalized by the number of classes. In addition to mAP, we also report average precision (AP) at specific intersection over union (IoU) thresholds: AP50 (IoU = 0.5) and AP75 (IoU = 0.75). Moreover, class-wise AP is provided to evaluate the detection performance for each object class individually.

B. Implementation Details

Experiments under the continual learning paradigm were conducted on an NVIDIA GeForce RTX 4090 GPU, with all methods implemented in PyTorch [43]. We adopted Faster R-CNN [44] as the primary detection network for multi-UAV cooperative maritime object recognition under the domain-incremental learning setting, using ResNet-50 [45] pretrained on ImageNet [46] as the backbone. Optimization was performed using stochastic gradient descent with a learning rate of 0.005 and a momentum of 0.9. The model was trained for 175 epochs in total with a batch size of 4.

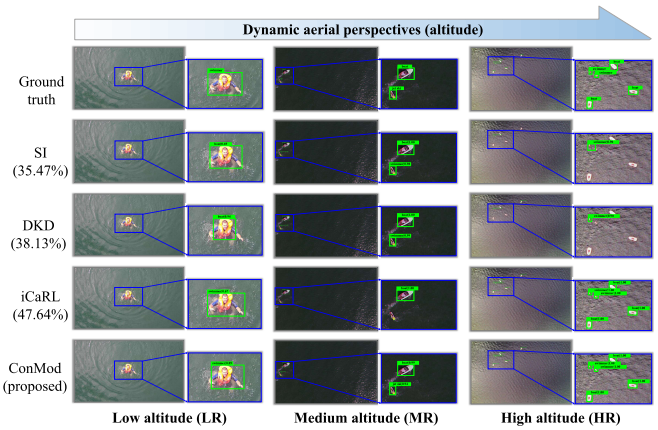


Fig. 7. Visualization of maritime object recognition results on the SeaDronesSee-CL dataset at different flight altitudes.

The tasks were arranged in descending order based on the corresponding mAP on the test set under the "individual training" strategy. Additionally, simple baseline experiments were conducted by incrementally fine-tuning a model across different aerial perspectives, referred to as the "naïve" approach. Comparison methods for multi-UAV cooperative maritime object recognition included: parameter regularization methods EWC [32] and SI [33]; functional regularization methods LwF [34], ERD [47], and DKD [48]; and replay-based methods DGR [37], iCaRL [49], A-GEM [36], ER [50], and LAPNet [30]. For parameter regularization methods, the regularization hyperparameters were set to $\lambda = 0.4$ for EWC and $\lambda = 0.0001$ for SI. For the functional regularization method, the distillation temperature was set to 2. Replay-based methods and the proposed ConMod framework were allocated a memory buffer capable of storing 70 examples. DGR employed a two-layer multilayer perceptron as its generator.

C. Experimental Results for Multi-UAV Cooperative Maritime Object Recognition on the SeaDronesSee-CL Dataset

Figs. 7 and 8 present the visualization results under dynamic aerial perspectives characterized by varying altitudes and angles. These results demonstrate that ConMod consistently outperforms state-of-the-art continual learning baselines, including parameter regularization method SI, function regularization method DKD, and replay-based method iCaRL, in multi-UAV cooperative maritime object recognition.

Specifically, Fig. 7 presents recognition performance across a range of UAV altitudes. In the LR perspective, ConMod effectively identifies swimmers with dynamic postures, while SI and DKD mistakenly classify swimmers as boats due to the relatively large object size at low altitudes. In the MR perspective, ConMod accurately recognizes both boats and jet skis. However, SI and DKD misclassify jet skis as swimmers due to their complex structures, and iCaRL suffers from missed detections of jet skis because it fails

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TABLE IV

QUANTITATIVE RESULTS OF THE PROPOSED CONMOD FRAMEWORK AND EXISTING CONTINUAL LEARNING METHODS ON THE SEADRONESSEE-CL DATASET. THE BEST RESULTS ARE IN RED, AND THE SECOND-BEST RESULTS ARE IN BLUE

Category	Methods	mAP (%)↑	AP50 (%)↑	AP75 (%)↑	Training time (h)↓	Inference efficiency (fps)↑
Naïve		36.92	66.20	35.14	5.20	8.04
Individual training		53.27	80.44	58.70	5.36	8.11
Parameter regularization	EWC [32]	34.92	61.09	33.82	5.82	8.13
	SI [33]	35.47	64.81	33.18	7.07	8.06
Functional regularization	LwF [34]	37.59	68.15	33.60	7.16	7.25
	ERD [47]	37.25	66.73	34.41	8.33	7.36
	DKD [48]	38.13	69.05	35.72	7.53	7.45
	DGR [37]	38.09	66.93	38.10	9.95	7.12
	DGR-Instruct [37]	47.22	77.29	48.85	10.32	7.04
Replay	iCaRL [49]	47.64	77.35	46.58	10.59	6.90
	A-GEM [36]	39.76	70.81	36.29	10.25	7.63
	ER [50]	47.43	76.42	51.11	10.03	7.44
	LAPNet [30]	47.34	76.84	48.19	10.37	7.55
Generative perspective-robust conflict estimation (proposed)		48.03	77.21	49.91	10.16	7.47
ConMod (proposed)		50.24	77.49	54.06	10.27	7.38

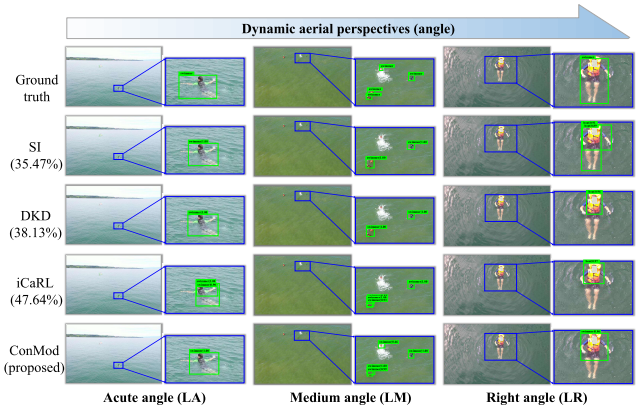


Fig. 8. Visualization of maritime object recognition results on the SeaDronesSee-CL dataset under varying angles.

to retain discriminative information during the knowledge accumulation process. Under the HR perspective, ConMod accurately localizes small-scale and densely packed maritime objects, benefiting from its robustness in preserving cross-perspective knowledge consistency. In contrast, EWC, SI, and DKD suffer from missed detections due to weak foreground signals. Fig. 8 illustrates recognition results under different aerial observation angles. Under the LA perspective, ConMod demonstrates notable robustness in handling acute observation angles and achieves precise object recognition. In comparison, iCaRL suffers from misdetections and false alarms. Under the LM perspective, ConMod exhibits strong discrimination by accurately separating maritime objects from background noise. In contrast, SI, DKD, and iCaRL fail to distinguish maritime objects from wave texture interference. In the LR perspective, ConMod demonstrates superior adaptability by adapting cross-perspective knowledge conflicts. Conversely, SI exhibits overlapping bounding boxes due to interference from surface and subsurface signals. Overall, these results highlight that ConMod retains perspective-robust knowledge and achieves accurate recognition across diverse UAV perspectives.

Table IV shows that the proposed ConMod framework achieves superior performance in multi-UAV cooperative maritime object recognition, with an mAP of 50.24%, AP50 of 77.49%, and AP75 of 54.06%. Specifically, ConMod outperforms the state-of-the-art iCaRL method by 2.6% in the mAP metric, demonstrating its stable optimization capability. In addition, ConMod surpasses ER and LAPNet methods by 2.95% and 5.87% in terms of the AP75 metric, indicating its superiority in achieving high-quality localization. By contrast, ConMod slightly underperforms the "individual training" framework (mAP: 53.27%) due to out-of-distribution noise perturbations. Existing continual learning methods such as EWC and SI exhibit limited capability for effective knowledge retention, yielding suboptimal results with mAPs of 34.92% and 35.47%, respectively. Other representative methods, including LwF, ERD, and DKD, show moderate improvements with mAPs of 37.59%, 37.25%, and 38.13%, respectively, but still exhibit limitations in maintaining discriminative representations. DGR employs a two-layer multilayer perceptron as the generator, whereas DGR-Instruct leverages a pretrained InstructDiffusion model. Experimental results indicate that DGR-Instruct achieves 47.22% mAP compared with 38.09% for DGR, demonstrating the superior generative capacity of the pretrained InstructDiffusion model. In terms of computational efficiency, ConMod achieves a training time of 10.27 h and an inference efficiency of 7.38 fps, which is superior to the state-of-the-art iCaRL method while remaining comparable to mainstream continual learning approaches. These results indicate that ConMod achieves a favorable balance between accuracy and efficiency for multi-UAV cooperative maritime object recognition by effectively identifying and mitigating cross-perspective knowledge conflicts.

Table V summarizes the experimental results of iCaRL and LAPNet across various network architectures, including YOLOv8-n [51], YOLOv8-s [51], RetinaNet [52], and Faster R-CNN with multiple backbones. Results indicate that Faster

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TABLE V
QUANTITATIVE RESULTS OF DIFFERENT NETWORKS FOR ICARL AND LAPNET ON THE SEADRONESSEE-CL DATASET

Method	Network	mAP (%)	AP50 (%)	AP75 (%)
iCaRL	YOLOv8-n	45.32	67.14	**49.68**
	YOLOv8-s	43.76	65.30	46.45
	RetinaNet (ResNet-50)	39.82	66.32	39.30
	Faster R-CNN (MobileNet)	27.71	52.41	24.05
	Faster R-CNN (ResNet-34)	46.74	**79.41**	45.83
	Faster R-CNN (ResNet-50)	**47.64**	77.35	46.58
	Faster R-CNN (ResNet-101)	44.89	74.92	44.46
LAPNet	YOLOv8-n	43.14	64.92	48.08
	YOLOv8-s	43.51	66.02	47.59
	RetinaNet (ResNet-50)	43.13	72.09	41.70
	Faster R-CNN (MobileNet)	31.80	58.64	30.28
	Faster R-CNN (ResNet-34)	45.95	**77.97**	46.03
	Faster R-CNN (ResNet-50)	**47.34**	76.84	**48.19**
	Faster R-CNN (ResNet-101)	43.92	72.40	46.05

TABLE VI
COMPARATIVE RESULTS ON THE SEADRONESSEE-CL DATASET. CLASSES 1–5 REPRESENT SWIMMER, BOAT, JET SKI, LIFE-SAVING APPLIANCES, AND BUOY, RESPECTIVELY. THE BEST RESULTS ARE IN BOLD

Method	AP per Class (%)					Average
	1	2	3	4	5	
EWC	19.98	59.62	21.06	7.64	66.30	34.92
SI	20.15	58.94	25.37	8.64	64.23	35.47
LwF	20.67	57.34	26.35	17.43	66.17	37.59
ERD	20.48	53.76	25.78	17.86	68.39	37.25
DKD	22.62	56.44	28.10	19.23	64.27	38.13
DGR	17.81	61.52	29.40	21.26	60.44	38.09
iCaRL	23.38	69.29	53.60	20.42	71.52	47.64
A-GEM	23.23	64.11	30.63	12.18	68.65	39.76
ER	23.35	68.14	51.02	21.22	**73.41**	47.43
LAPNet	24.86	69.73	55.15	21.51	65.46	47.34
ConMod (proposed)	**25.19**	**69.93**	**59.99**	**22.72**	73.37	**50.24**

R-CNN with ResNet-50 achieves the highest performance, attaining mAP values of 47.64% and 47.34% for iCaRL and LAPNet, respectively. YOLOv8-n and YOLOv8-s exhibit relatively superior performance compared to RetinaNet, which can be attributed to architectural enhancements, including a decoupled detection head that improves the balance between classification and localization, and multiscale feature aggregation that enhances feature robustness. In the backbone comparison, ResNet-34 and ResNet-50 achieve superior performance relative to ResNet-101, owing to overfitting induced by the increased complexity of deeper networks.

Table VI presents the performance of the proposed ConMod framework across five maritime object categories, demonstrating its effectiveness in mitigating cross-perspective conflicts through accumulated representational knowledge. ConMod achieves competitive AP metrics for swimmer (25.19%), boat (69.93%), jet ski (59.99%), life-saving appliances (22.72%), and buoy (73.37%). Notably, it yields a 6.39% AP improvement over the state-of-the-art iCaRL method on the jet ski category, attributed to its ability to accurately identify maritime objects under dynamic aerial perspectives and to suppress missed detections. For the buoy category, both ConMod and ER achieve high AP of 73.37% and 73.41%, benefiting from the distinctive visual characteristics (e.g., color and shape) of buoys that support reliable feature representation. Among existing continual learning methods, iCaRL, ER, and LAPNet demonstrate comparatively strong performance relative to other approaches, as they explicitly preserve representative experiences to maintain cross-perspective knowledge, contributing to improved recognition performance across multiple maritime object categories. However, continual learning methods such as SI and DKD exhibit inferior performance in multi-UAV cooperative maritime object recognition across diverse object categories.

Fig. 9 illustrates that the proposed ConMod framework achieves stable mAP performance across different UAV perspectives for sequentially increasing tasks. Specifically, the tasks are arranged in descending order based on their mAP on the test set, with task indices 1, 6, 3, 5, 2, 7, and 4 corresponding to HR, LAM, LR, LM, MR, LA, and LMR, respectively. Compared to the ER method, ConMod achieves mAP improvements of 5.38%, 7.26%, 2.16%, and 8.55% under the perspectives of LM, MR, and LA, LMR, respectively. This indicates that ConMod continually accumulates knowledge as the cross-perspective scenarios expand, ensuring consistent improvements in multi-UAV cooperative maritime object recognition. Nevertheless, the ER method exhibits inferior performance under the LA and LMR perspectives, with mAP scores of 25.08% and 38.99%, respectively. This suggests that ER cannot aggregate knowledge from diverse observation angles. Overall, the proposed ConMod framework demonstrates stable cross-perspective continual learning performance, while the ER method exhibits substantial fluctuations in mAP performance, revealing that ConMod effectively addresses conflict management and knowledge forgetting.

D. Experimental Results for Multi-UAV Cooperative Maritime Object Recognition on the SeaDronesSee-CL-v2 Dataset

Fig. 10 illustrates the capability of the proposed ConMod framework to accurately recall small maritime objects with high confidence on the SeaDronesSee-CL-v2 dataset. In the LR perspective, ConMod detects a floater with a confidence score of 0.99, outperforming iCaRL (0.93). In the MR perspective, ConMod accurately distinguishes two densely distributed small objects, whereas DKD and iCaRL exhibit missed detections. From the HR perspective, ConMod effectively localizes multiple maritime objects in complex marine scenarios, while iCaRL fails to detect the small floater.

Table VII demonstrates that the proposed ConMod framework achieves the best performance, with an mAP of 60.7% and an AP50 of 90.99%, surpassing iCaRL (mAP of 57.84% and AP50 of 86.83%). This improvement is attributed to the conflict modulation mechanism, which effectively addresses variations in observation altitudes under nadir perspectives. Furthermore, the ER method attains the highest AP75 (61.46%) among existing continual learning approaches, highlighting strong localization capability. This advantage

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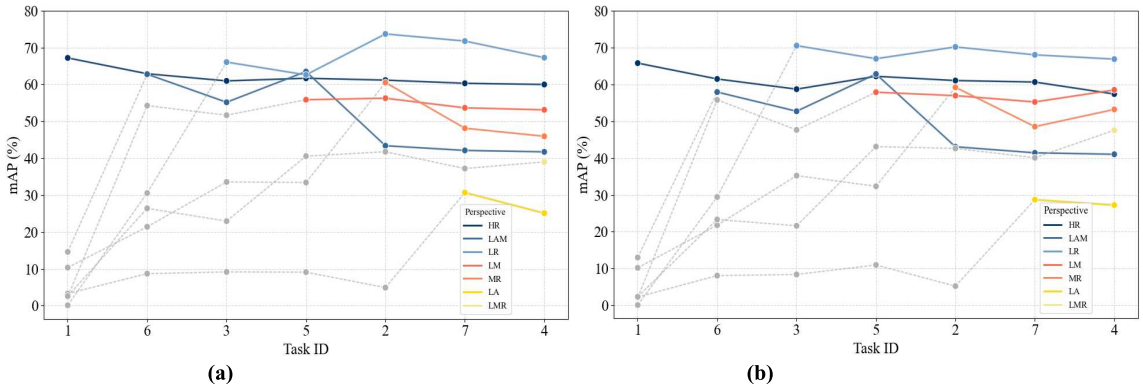


Fig. 9. Performance of baseline and ConMod methods across different UAV perspectives. Gray and colored lines represent evaluations conducted before and after training on the respective tasks. (a) Baseline (ER). (b) ConMod (proposed)

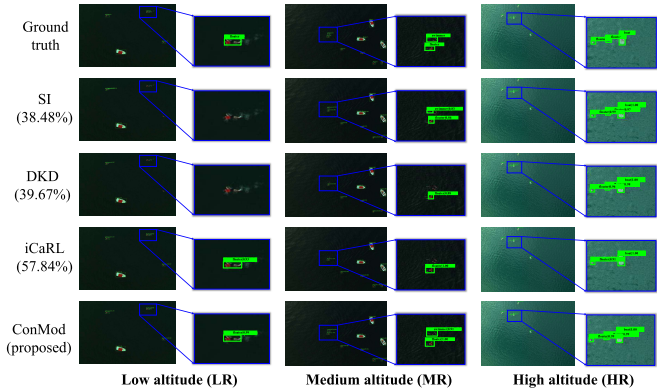


Fig. 10. Visualization results on the SeaDronesSee-CL-v2 dataset under nadir perspectives at multiple altitudes.

TABLE VII

QUANTITATIVE RESULTS ON THE SEADRONESSEE-CL-V2 DATASET, WITH THE BEST RESULTS IN RED AND SECOND BEST IN BLUE

Category	Methods	mAP (%)	AP50 (%)	AP75 (%)
Naïve		38.34	57.57	38.18
Individual training		57.29	90.31	57.04
Parameter regularization	EWC [32]	37.75	57.89	38.57
	SI [33]	38.48	57.78	40.60
Functional regularization	LwF [34]	39.10	59.22	40.48
	ERD [47]	37.26	56.15	38.51
	DKD [48]	39.67	61.13	39.72
Replay	DGR [37]	50.30	84.21	48.98
	DGR-Instruct [37]	56.43	86.26	56.81
	iCaRL [49]	57.84	86.83	59.54
	A-GEM [36]	48.06	76.26	44.73
	ER [50]	56.62	86.07	61.46
	LAPNet [30]	57.20	86.51	53.51
ConMod (proposed)		60.70	90.99	61.09

arises from the design of ER as a general replay-based strategy, enabling rapid adaptation for object bounding box refinement.

Evaluation on the two benchmark datasets indicates that ConMod achieves consistent performance improvements across varying and fixed observation angles.

IV. DISCUSSION

A. Ablation Study of ConMod Framework

In Table VIII, components 1–4 denote: 1) generative perspective-robust conflict estimation; 2) perspective-specific triplet loss; 3) ConMod loss derived from the environmental variation branch; and 4) conflict-modulated loss derived from the geometric distortion branch, with components 2–4 collectively constituting the ConMod continual learning module. The results indicate that ConMod, integrating all the components, achieves the highest performance, with 50.24% in mAP. These improvements are mainly attributed to generative perspective-robust conflict estimation (+ 0.6% mAP), perspective-specific triplet loss (+0.66% mAP), and the ConMod losses (+0.7% mAP from the environmental variation branch and +0.85% mAP from the geometric distortion branch).

TABLE VIII

ABLATION STUDY ON THE INDIVIDUAL COMPONENTS OF CONMOD. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

1	2	3	4	mAP (%)	AP50 (%)	AP75 (%)
				47.43	76.42	51.11
✓				48.03	77.21	49.91
✓	✓			48.69	**77.58**	51.91
✓	✓	✓		49.39	76.94	53.70
✓	✓	✓	✓	**50.24**	77.49	**54.06**

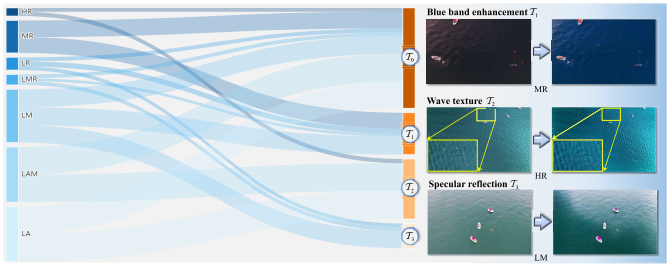


Fig. 11. Sankey diagram and visualization illustrate the effects of perspective-constrained image editing instructions ($\mathcal{T}_0$–$\mathcal{T}_3$), corresponding to no editing, blue band enhancement, wave texture, and specular reflection, respectively.

B. Ablation Study on Generative Perspective-Robust Conflict Estimation for Cross-Perspective Conflict Representation

Fig. 11 presents the Sankey diagram and image editing visualization results of the perspective-constrained image editing

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TABLE IX

ABLATION RESULTS FOR RADIATION DISTORTION CORRECTION AND IMAGE EDITING INSTRUCTIONS ON THE PERSPECTIVE-AWARE SCENE GENERATOR. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Radiation correction / Image editing	mAP (%)	AP50 (%)	AP75 (%)
No correction	47.43	76.42	**51.11**
Histogram equalization	46.09	74.43	46.60
Contrast adjustment	46.22	**77.81**	46.86
Brightness adjustment	46.49	77.40	47.48
Gamma correction	47.10	76.65	49.40
Random image editing	46.25	73.97	49.89
Image editing instruction $\mathcal{T}_1$	47.76	73.77	48.76
Image editing instruction $\mathcal{T}_2$	46.77	71.00	46.58
Image editing instruction $\mathcal{T}_3$	46.80	71.47	46.31
Perspective-constrained (proposed)	**48.03**	77.21	49.91

instructions $\mathcal{T}_1$, $\mathcal{T}_2$, and $\mathcal{T}_3$. Specifically, $\mathcal{T}_1$ is primarily applied to the MR and LM perspectives to simulate data distribution shifts caused by UAV altitude variations from medium or low to high. $\mathcal{T}_2$ mainly affects perspectives such as LAM, LA, and HR, while $\mathcal{T}_3$ is mainly applied to LM, capturing diverse data distribution variations under dynamic aerial perspectives. In detail, the perspective-aware scene generator explicitly applies blue band enhancement $\mathcal{T}_1$ to simulate Rayleigh scattering effects associated with high-altitude UAV perspectives. In addition, it introduces wave textures $\mathcal{T}_2$ to enhance wave perturbations by incorporating high-frequency patterns into the sea surface. The generator further utilizes the image editing instruction $\mathcal{T}_3$ associated with specular reflection to simulate dynamic brightness variations on the water surface, capturing the angle-dependent reflectance characteristics under varying observation perspectives.

Table IX presents the results under different radiation distortion correction and image editing settings, with histogram equalization applied globally across the entire image, contrast adjustment performed with an enhancement factor of 1.2, brightness adjustment with an enhancement factor of 1.05, and gamma correction with a correction factor of 1.2. Specifically, radiation distortion correction algorithms yield no improvement in recognition accuracy relative to the baseline (no correction), as they primarily perform global radiometric normalization, which is insufficient to account for the dynamic variations in maritime scenes across different UAV perspectives. Gamma correction achieves comparatively superior performance (mAP 47.1%), as it applies nonlinear scaling that effectively preserves subtle feature variations essential for robust maritime object recognition. Compared to random image editing, the proposed method yields improvements of 1.78% in mAP and 3.24% in AP50, indicating that incorporating maritime knowledge priors through the probabilistic state transition graph effectively captures the global contextual relationship between UAV perspective variations and cross-perspective data distribution shifts. However, the improvement in AP75 is only marginal (0.02%) over random image editing, indicating that the proposed perspective-aware scene generator primarily simulates global data distribution shifts rather than enhancing the spatial localization accuracy of individual maritime objects. Comparing individual image editing instructions $\mathcal{T}_1$–$\mathcal{T}_3$, $\mathcal{T}_1$ achieves the highest performance with an mAP of 47.76%. This is attributed to the use of blue band enhancement, which introduces Rayleigh scattering to increase spectral diversity, thereby improving the generalization capability across varying UAV perspectives.

TABLE X

ABLATION RESULTS FOR DIFFERENT CONFLICT INDICATOR ESTIMATION BRANCHES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Image (Env.)	Text (Geo.)	mAP (%)	AP50 (%)	AP75 (%)
✓		49.39	76.94	53.70
	✓	49.56	76.77	54.01
✓	✓	**50.24**	**77.49**	**54.06**

Table X presents the ablation study on the environmental variation branch (incorporating image information) and the geometric distortion branch (incorporating textual information). The results indicate that the integration of both branches yields superior performance, achieving an mAP of 50.24% due to the complementary contributions of image and text modalities. Individually, the geometric distortion branch attains an mAP of 49.56%, as UAV perspective metadata provides geometric priors that mitigate noise arising from visual variations, thereby supporting robust conflict estimation. The environmental variation branch achieves a higher AP50 of 76.94%, as image information captures contextual relationships that facilitate the modeling of discriminative cross-perspective conflicts.

Table XI presents a comparative analysis of various conflict index estimation strategies for generative perspective-robust conflict estimation. The channel dimensions involved in each strategy are indicated in parentheses. For instance, $128 \times 2 \rightarrow 256 \rightarrow 64$ denotes that two 128-D feature vectors extracted from separate branches are first concatenated into a 256-D feature representation, which is subsequently projected to a 64-D feature vector through a fully connected layer. Experimental results demonstrate that the proposed dual-modal conflict index estimator achieves superior performance in multi-UAV cooperative maritime object recognition. This improvement arises from the parallel dual-modal encoding mechanism, which reduces the adverse effects of the distribution gap between geometric and environmental features, enabling accurate conflict index estimation under dynamic aerial perspectives. Specifically, this design facilitates efficient end-to-end learning and achieves superior recognition performance, with an mAP of 50.24%, an AP50 of 77.49%, and an AP75 of 54.06%. Given ten independent trials, the proposed dual-modal conflict index estimator achieves a mean mAP improvement of approximately 0.59% over the feature concatenation strategy ($128 \times 264 \times 212864$), exhibiting minimal variability and indicating that the observed gain is statistically robust.

C. Ablation Study on ConMod Continual Learning for Globally Optimal Parameter Optimization

Table XII summarizes the ablation study on the weight coefficient λ_{PS} in the ConMod continual learning module.

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TABLE XI

ABLATION RESULTS FOR CONFLICT INDEX ESTIMATION COMPARING FEATURE CONCATENATION AND THE PROPOSED DUAL-MODAL CONFLICT INDEX ESTIMATOR. THE VALUES FOLLOWING EACH RESULT INDICATE MEAN ± STANDARD DEVIATION OVER TEN RUNS. THE BEST RESULTS ARE IN BOLD

Conflict index estimation	**mAP (%) [mean ± std]**	**AP50 (%) [mean ± std]**	**AP75 (%) [mean ± std]**
Feature concatenation (128×2→256→64)	47.92 (47.95 ± 0.12)	76.97 (77.05 ± 0.28)	53.27 (53.30 ± 0.25)
Feature concatenation (128×2→256→128→64)	47.04 (47.10 ± 0.15)	76.04 (76.10 ± 0.30)	51.75 (51.80 ± 0.20)
Feature concatenation (128×2→64×2→128→64)	49.78 (49.66 ± 0.14)	76.45 (76.50 ± 0.26)	53.71 (53.75 ± 0.22)
Dual-modal conflict index estimator (128×2→64×2)	**50.24 (50.25 ± 0.09)**	**77.49 (77.44 ± 0.16)**	**54.06 (54.23 ± 0.19)**

TABLE XII

ABLATION RESULTS IN THE WEIGHT COEFFICIENT λ_{PS} IN THE CONMOD CONTINUAL LEARNING MODULE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Weight coefficient λ_{PS}	**mAP (%)**	**AP50 (%)**	**AP75 (%)**
0.5	**50.24**	**77.49**	**54.06**
0.75	49.47	76.83	52.52
1	49.56	76.80	52.65
1.25	49.19	76.39	52.97
1.5	49.43	76.11	54.05

The results demonstrate that the proposed ConMod continual learning module achieves optimal performance when λ_{PS} = 0.5, attaining an mAP of 50.24%. This improvement stems from the perspective-specific triplet loss, which acts as a regularizer to align geometric and environmental features. A moderate weight effectively adjusts model parameters while preserving balance in the overall optimization. In contrast, large values of λ_{PS} hinder stable convergence, as the indirect optimization objective may introduce biased gradients, leading to suboptimal performance. Accordingly, λ_{PS} = 0.5 is adopted as the optimal setting for the ConMod continual learning module.

Fig. 12 exhibits the probability distribution of dual-modal feature distances between geometric features extracted from the geometric distortion branch and environmental features derived from the environmental variation branch. The blue and orange areas represent the feature distributions with and without the perspective-specific triplet loss, respectively. The results indicate that the perspective-specific triplet loss significantly reduces dual-modal feature distances. This loss explicitly enforces adaptive constraints on cross-perspective feature discrepancies, promoting a compact and discriminative feature distribution aligned with UAV perspective variations. In contrast, the absence of this constraint leads to dispersed feature distributions, primarily due to the lack of explicit feature clustering mechanisms, which impedes the effective alignment of geometric and environmental features within a unified embedding space. Furthermore, the perspective-specific triplet loss facilitates the compression of the feature distribution with reduced variance, thereby contributing to the construction of perspective-invariant representations and enhancing the model's robustness. When the perspective-specific triplet loss is not employed, UAV data captured across

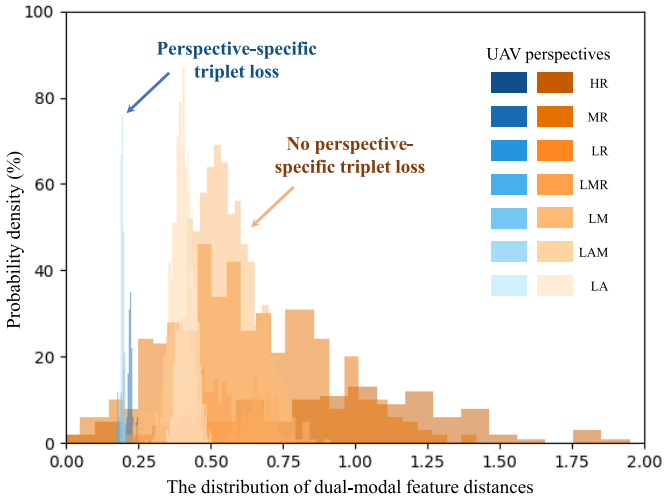


Fig. 12. Dual-modal feature distance distribution illustrates the effectiveness of the perspective-specific triplet loss in enhancing feature consistency across UAV perspectives.

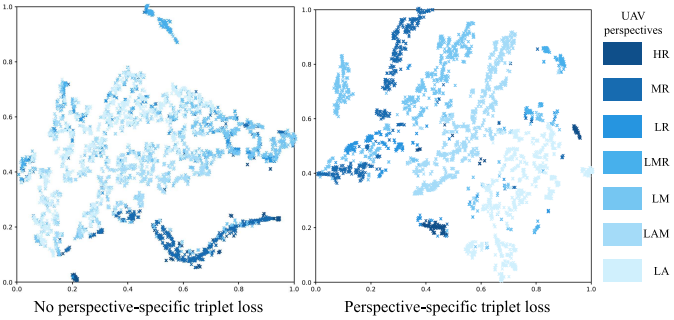


Fig. 13. t-SNE visualization results reveal that the proposed perspective-specific triplet loss enhances perspective-aware discriminative feature representations.

different altitudes (HR, MR, and LR) tend to exhibit highly variable and unstable feature distributions, due to the inherent difficulty in learning scale-invariant features. By contrast, UAV data collected across different observation angles (LM, LAM, and LA) at a fixed altitude demonstrate relatively consistent feature distances since angular variations exhibit lower representational ambiguity under uniform scale conditions.

Fig. 13 shows that the proposed perspective-specific triplet loss enhances perspective-aware discriminative features, as reflected by the clustering of environmental features by specific UAV perspectives. Specifically, the t-SNE was employed

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TABLE XIII

ABLATION OF NEGATIVE SAMPLING IN PERSPECTIVE-SPECIFIC TRIPLET LOSS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Sampling strategy	mAP (%)	AP50 (%)	AP75 (%)	Training time (h)
Random sampling	**50.24**	77.49	**54.06**	10.27
Hard-negative mining	49.03	**77.54**	53.02	10.33

TABLE XIV

ABLATION RESULTS OF THE CONMOD LOSS WITH VARYING α AND β, BALANCING GEOMETRIC AND ENVIRONMENTAL CONTRIBUTIONS TO CONFLICT INDEX C_i. THE BEST RESULTS ARE IN BOLD

α	β	mAP (%)	AP50 (%)	AP75 (%)
0	1	49.39	76.94	53.70
0.25	0.75	49.82	76.98	53.09
0.5	**0.5**	**50.24**	**77.49**	**54.06**
0.75	0.25	49.17	76.80	53.48
1	0	49.56	76.77	54.01

for dimensionality reduction, with principal component analysis employed as a preprocessing step. An early exaggeration parameter of 30 was adopted to facilitate the separation of data points corresponding to different UAV perspectives. The proposed perspective-specific triplet loss demonstrates a strong capacity to distinguish different observation angles, as evidenced by the well-separated feature distributions corresponding to the LM, LAM, and LA perspectives. Additionally, it exhibits favorable clustering performance concerning UAV altitude, with HR, MR, and LR perspectives forming distinct and compact feature clusters. In the absence of the perspective-specific triplet loss, the feature distributions associated with different UAV perspectives exhibit significant overlap, resulting in diminished interperspective separability. Notably, the features corresponding to the HR and MR perspectives present distinct distribution patterns that deviate from those of the remaining perspectives, further highlighting the heterogeneity in perspective-specific feature representations.

Table XIII compares the proposed random sampling strategy with the hard-negative mining strategy in perspective-specific triplet loss. The hard-negative mining strategy achieves a slightly lower mAP of 49.03% than the random strategy, as it relies on highly similar negative samples derived from limited UAV perspectives, which reduces diversity and increases the risk of overfitting. In addition, the hard-negative mining strategy is less computationally efficient and requires 10.33 h of training. Overall, the random strategy achieves competitive performance in both accuracy and efficiency compared with the hard-negative mining strategy.

Table XIV presents the ablation results of the ConMod loss with different hyperparameters α and β. When both parameters are set to 0.5, it achieves the best recognition performance, with an mAP of 50.24%, an AP50 of 77.49%, and an AP75 of 54.06%. These results suggest that a balanced weighting of the geometric distortion and environmental variation branches enables accurate conflict index estimation, facilitating effective optimization of the ConMod loss. The geometric distortion branch ($\alpha = 1$ and $\beta = 0$) and the environmental variation branch ($\alpha = 0$ and $\beta = 1$) make equally important contributions to the estimation of the conflict index C_i in the ConMod loss, as they capture discrepancies from UAV perspective distortions and maritime environmental variations, jointly supporting comprehensive conflict estimation. When a higher weight is assigned to the environmental variation branch ($\alpha = 0.25$ and $\beta = 0.75$), the model achieves a slightly higher mAP of 49.82% compared to the setting favoring the geometric distortion branch ($\alpha = 0.75$ and $\beta = 0.25$). This result indicates that the environmental variation branch offers a more global characterization of conflict index estimation, facilitating robust performance improvements compared to the geometric variation branch.

V. CONCLUSION

In this study, the proposed ConMod framework, which integrates generative perspective-robust conflict estimation with ConMod continual learning, is introduced for multi-UAV cooperative maritime object recognition. This framework introduces a conflict index to quantify the data distribution shifts induced by cross-perspective inconsistencies and utilizes this index to learn systematic distribution shift patterns across multiperspective maritime data. Experimental results on the SeaDronesSee-CL dataset demonstrate that ConMod outperforms the state-of-the-art iCaRL method by 2.6% in terms of mAP, indicating its robust capability in cross-perspective conflict management and knowledge association. Ablation results further reveal that the generative perspective-robust conflict estimation constructs a comprehensive conflict index to represent cross-perspective discrepancies, while ConMod continual learning effectively identifies discriminative conflict experiences, contributing to efficient parameter optimization and accurate multi-UAV cooperative maritime object recognition. Future research will investigate incorporating frequency-domain representations into continual learning to improve the discriminative capacity and adaptability of fine-grained maritime object recognition under dynamic aerial perspectives, with a particular focus on leveraging sea surface frequency characteristics to capture subtle environmental variations.

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